

PALMYRA, Syrian Arab Republic

WMO: 400610

Lat: 34.557N

Lon: 38.317E

Elev: 1322

StdP: 14.01

Time Zone: 2.00 (SYR)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.4	33.8	11.8	10.5	47.6	15.5	12.7	48.1	17.9	48.1	14.1	47.5	1.4	270	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.5	105.7	68.5	103.1	67.6	100.7	67.1	72.9	93.3	71.3	91.8	70.1	90.4	3.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	105.0	79.8	65.6	99.4	79.1	64.3	94.7	78.4	37.5	92.7	36.0	91.4	34.9	90.3	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
18.0	15.2	12.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			27.3	110.7	4.0	3.9	24.4	113.5	22.1	115.8	19.8	118.0	16.9	120.8
			24.1	77.1	3.2	3.5	21.7	79.7	19.9	81.7	18.1	83.6	15.7	86.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	67.7	45.6	50.2	57.2	66.1	75.6	83.1	87.5	87.6	81.3	72.0	57.2	48.0	(6)
(7)		DBStd	15.89	4.64	5.22	6.53	6.60	5.98	4.53	3.74	3.64	4.11	5.65	5.91	5.13	(7)
(8)		HDD50	328	147	56	13	1	0	0	0	0	0	13	98	(8)	
(9)		HDD65	2113	601	415	256	62	2	0	0	0	11	238	528	(9)	
(10)		CDD50	6792	11	61	236	483	795	992	1162	1166	939	682	230	35	(10)
(11)		CDD65	3101	0	0	14	94	332	542	697	701	489	228	4	0	(11)
(12)		CDH74	38440	0	0	134	902	3701	7109	9763	9646	5396	1764	25	0	(12)
(13)		CDH80	22014	0	0	33	293	1792	4194	6194	6052	2862	593	1	0	(13)
(14)	Wind	WSAvg	4.0	2.2	2.9	3.9	4.6	5.0	5.7	6.8	6.0	4.4	2.6	1.9	1.9	(14)
(15)	Precipitation	PrecAvg	5.1	0.8	0.8	0.9	0.6	0.4	0.0	0.0	0.0	0.4	0.6	0.8	(15)	
(16)		PrecMax	10.6	2.6	5.3	4.3	3.3	2.6	0.6	0.0	0.1	1.7	2.5	3.3	(16)	
(17)		PrecMin	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	(17)	
(18)		PrecStd	2.2	0.6	0.9	0.9	0.8	0.5	0.1	0.0	0.0	0.4	0.7	0.6	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.6	70.4	84.3	94.3	99.9	105.6	110.4	108.9	101.7	93.6	77.0	64.9	(19)
(20)		MCWB	50.9	52.7	56.7	61.6	65.9	67.0	68.0	70.0	68.3	65.6	57.9	53.3	(20)	
(21)		2%	DB	58.7	65.5	77.3	87.7	95.7	102.0	106.3	105.7	98.3	89.5	73.1	60.9	(21)
(22)		MCWB	49.0	50.3	54.2	59.4	63.4	66.5	67.7	69.6	66.8	63.1	56.2	51.2	(22)	
(23)		5%	DB	56.4	62.8	72.9	83.5	93.0	99.4	103.4	103.0	95.6	86.3	70.2	58.5	(23)
(24)		MCWB	47.9	49.6	52.9	58.2	62.4	65.5	67.4	68.1	65.9	62.2	55.4	49.9	(24)	
(25)		10%	DB	53.9	60.0	68.9	79.4	89.9	96.9	100.7	100.5	93.0	83.1	67.3	56.2	(25)
(26)		MCWB	46.6	48.4	51.7	56.7	61.7	64.6	67.1	67.6	65.1	61.1	54.1	48.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	54.1	59.6	66.6	71.1	72.0	75.3	75.2	72.9	69.9	62.4	55.0	(27)
(28)		MCDWB	59.2	65.4	73.1	84.4	90.1	97.7	97.9	97.2	92.5	85.8	70.8	59.7	(28)	
(29)		2%	WB	50.5	52.0	57.1	62.0	66.8	69.2	72.1	73.1	70.5	66.3	60.0	53.1	(29)
(30)		MCDWB	56.5	62.6	70.6	80.3	87.3	93.1	94.3	92.8	88.8	80.8	68.2	58.1	(30)	
(31)		5%	WB	49.1	50.7	55.1	59.8	64.8	67.4	70.5	71.6	68.9	64.6	58.0	51.6	(31)
(32)		MCDWB	54.5	60.4	69.2	78.9	85.8	90.7	92.6	91.3	85.7	79.7	66.0	56.3	(32)	
(33)		10%	WB	47.6	49.4	53.2	58.1	63.2	66.2	69.4	70.5	67.7	63.3	56.1	50.1	(33)
(34)		MCDWB	52.6	58.1	66.6	76.7	85.0	91.0	91.1	89.9	83.8	78.2	64.4	54.9	(34)	
(35)	Mean Daily Temperature Range	MDBR	14.7	17.1	19.0	21.5	24.0	25.6	26.5	25.9	23.2	20.5	19.0	15.2	(35)	
(36)		5% DB	MCDBR	17.9	20.7	24.1	25.9	27.1	27.9	28.8	27.6	26.1	24.1	22.2	18.5	(36)
(37)		MCWBR	10.1	9.9	9.6	9.8	9.2	7.7	6.4	6.0	7.7	9.3	11.1	10.7	(37)	
(38)		5% WB	MCDBR	15.5	18.1	20.2	23.1	24.5	26.9	26.9	26.6	23.8	20.9	17.0	14.9	(38)
(39)		MCWBR	9.5	9.2	9.1	10.0	9.6	8.0	7.0	6.6	7.5	8.4	9.4	9.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.363	0.396	0.435	0.499	0.496	0.432	0.426	0.432	0.422	0.454	0.390	0.370	(40)	
(41)		taud	2.145	2.044	1.936	1.789	1.826	2.015	2.047	2.027	2.041	1.943	2.087	2.122	(41)	
(42)		Ebn at Noon	259	263	263	252	255	271	272	268	264	240	246	247	(42)	
(43)		Edh at Noon	38	47	56	69	67	56	53	53	50	51	40	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	897	1182	1613	1961	2284	2601	2500	2232	1886	1405	1040	819	(44)	
(45)		RadStd	45	89	66	85	112	61	52	72	72	73	72	59	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page